

Seihwan Jung

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Education

Seoul National University (SNU) Mar. 2024 – Feb. 2028 (expected)
B.S. in Mechanical Engineering *Seoul, South Korea*

- GPA: 3.55/4.3

Technische Universität Berlin (TU Berlin) Oct. 2026 – Mar. 2027
Exchange Student *Berlin, Germany*

- Winter Semester 2026/27; continuing coursework and research in robotic manipulation.

Research Interests

- **Grasping and Manipulation:** Passive compliance and adaptive grasp stability – letting a mechanism conform to an object instead of controlling every joint.
- **Robotic Hardware Design:** Underactuated, tendon-driven mechanisms, from CAD through fabrication and assembly to tested hardware.
- **Simulation-to-Real:** Using physics simulation sweeps to choose design parameters, then verifying against measurements on the physical build.

Experience

SNU BioRobotics Laboratory (BRL) Jan. 2026 – Aug. 2026
Undergraduate Researcher *Seoul, South Korea*

- Designed, fabricated, and validated a tendon-driven prosthetic hand with a fully passive abduction/adduction (ADD/ABD) degree of freedom.
- Full pipeline ownership: mechanical design, CAD (Fusion 360), MuJoCo simulation sweeps, 3D-printed fabrication, and physical testing.

Selected Projects (see more at sehwanjung.netlify.app)

Passive-ABD Tendon-Driven Prosthetic Hand [More Info.](#)

- Fully passive ADD/ABD joint at the MCP using two opposing torsion springs and dual-tendon routing – no extra motors or added weight.
- Selected spring stiffness via a MuJoCo simulation sweep across object families, then implemented physical control with Dynamixel actuators.

Honors / Awards

2nd Place – Creative Engineering Design Robocon *Nov. 2025*
Department of Mechanical Engineering 33rd Anniversary, Seoul National University

Skills

- **Programming:** Python, C++, MATLAB
- **Hardware:** CAD (Fusion 360), 3D Printing, Motor Control (Dynamixel), Microcontrollers (Arduino, Open-
RB, ESP-32), Workshop Fabrication
- **Simulation:** MuJoCo
- **Languages:** Korean (Native), English (Working Proficiency)